

HONOURS THESIS DEFENCE • COMP 499B

Programmatic Access to the Nature's Palette Spectral Data Repository via a REST API and R Interface

Tridib Paul Turjo

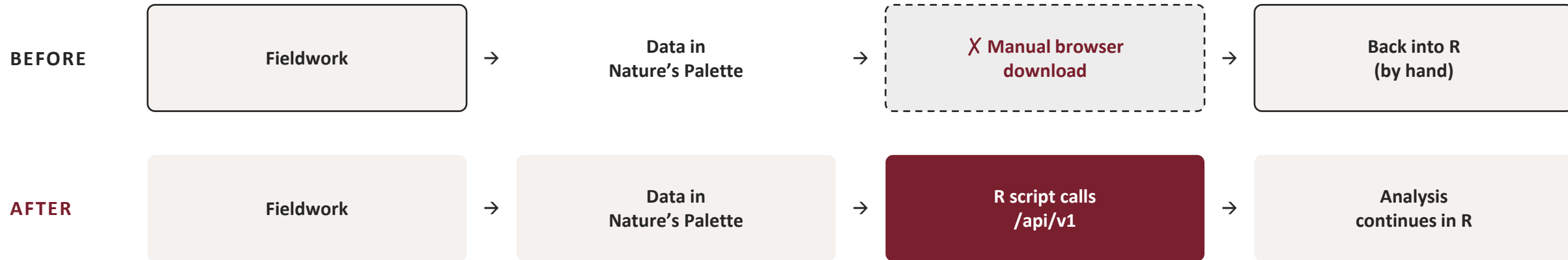
B.Sc. (Honours), Computer Science — Memorial University of Newfoundland

Supervised by Dr. Adrian Fiech and Dr. Pierre-Paul Bitton

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Why this matters



- Spectral data are costly to collect (fieldwork and equipment time make reuse valuable)
- Researchers analyze spectra in R (e.g. the pavo package)
- Nature's Palette was browser-only: not scriptable, not reproducible
- The break was always between the repository and R - this thesis closes that gap

What this thesis asks, and claims

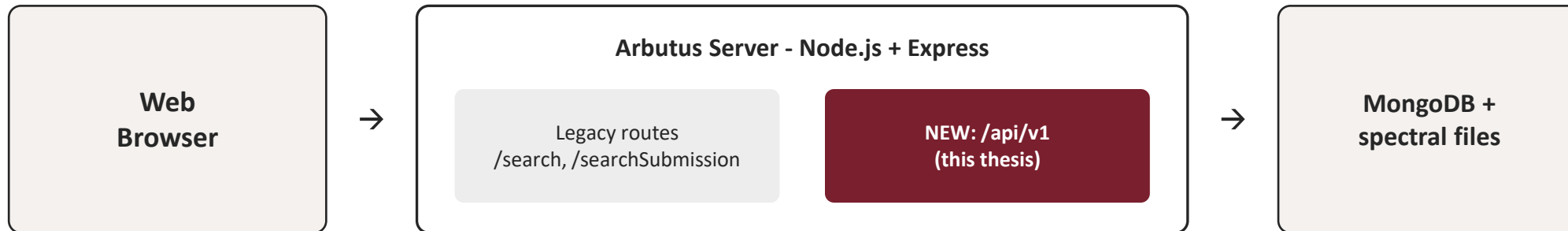
Research question How can Nature's Palette support reproducible, script-based search and download without breaking the live website?

Design claim A thin REST layer under `/api/v1` that reuses existing server logic, plus a thin R client, can match web behaviour for equivalent queries.

Objectives

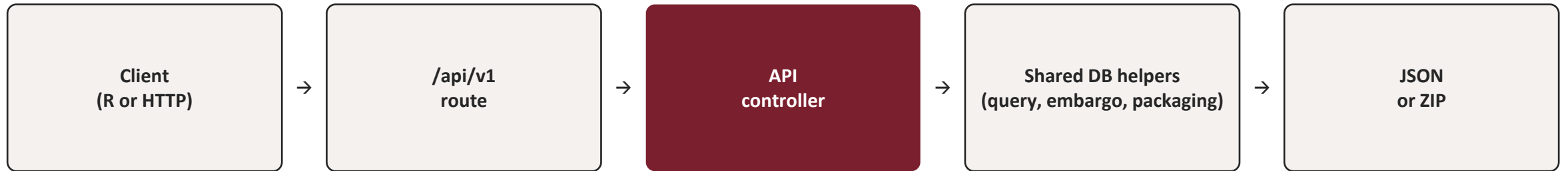
- REST API for catalogue / submission search, download, metadata
- R package `naturepaletteR` wrapping the API for scripted workflows
- Evaluate correctness, web/API/R parity, and indicative performance

Nature's Palette today, and what this thesis adds



- Node.js, Express, MongoDB - hosted on Compute Canada Arbutus
- Web already supports catalogue/submission search, embargo rules, ZIP download
- Prior honours work improved validation and colour-space search - not scripted access
- This thesis adds an additive `/api/v1` layer plus an R client

How a request actually flows through the system



- Every request - search or download - runs through the same query, grouping, and download helpers as the website
- Response type depends on the request: JSON for search and metadata, a binary ZIP for downloads
- For downloads specifically, the controller loads a stored SearchResult and routes to the catalogue or submission packaging pipeline

Four design principles

01

Mirror web behaviour

Same search, embargo, and packaging logic - no new query semantics

02

Additive, not disruptive

New routes under /api/v1; legacy routes and pages untouched

03

Modular code

controllers/api/ and routes/api/, separated from legacy modules

04

Thin R client

naturepaletteR wraps the API; it never reimplements server logic

Four endpoints under /api/v1

Method	Endpoint	Role
GET	/api/v1/catalogue/search	Specimen-level search
GET	/api/v1/submissions/search	Submission-level search
POST	/api/v1/download	ZIP archive by searchResultId
GET	/api/v1/datasets/{id}/metadata	Structured JSON metadata



- Reuses existing Mongoose models, embargo rules, and packaging logic (no new biology)
- Search first, then download: a client stores a SearchResult and later downloads by ID without re-querying

naturepaletteR: a thin R wrapper

```
> library(naturepaletteR)
> s <- np_search_catalogue(query = list(genus = "Trogon", sex = "male"))
> summary(s)
=== Nature Palette Search Summary ===

Search Type:
catalogue

Search Query:
$genus
[1] "Trogon"

$sex
[1] "male"

Total Matches: 4666
```

- Thin wrapper built on httr2 + jsonlite, with no server logic reimplemented
- One shared np_search S3 object, so summary() and np_download() work for both search types
- Polygon search: client validates coordinates and auto-computes the required mapCoordinate
- Installed from GitHub (remotes::install_github), not yet on CRAN

What a researcher can do now

1

Search in R

```
np_search_catalogue() or  
np_search_submissions()
```



2

Inspect

```
summary(s) - type, query, match  
count, preview
```



3

Download

```
np_download(s, path = "out") -  
same ZIP as the website
```

Same results as the website - inside a reproducible script.

Iterative development, weekly feedback



Two examples from the build

- Submission download routing: integration testing found POST /api/v1/download only invoked the catalogue handler — fixed by adding routing logic in the API layer
- Polygon search: added after supervisor feedback identified it as valuable; required client-side coordinate validation

Automated correctness checks

API TEST SUITE

4 / 4

groups passing: health, catalogue (incl. OR/NOT),
submissions, both download paths

R PACKAGE (testthat)

PASS 21

FAIL 0 · WARN 0 · SKIP 0, from offline polygon validation +
live integration checks

Automated correctness checks, not a user study.

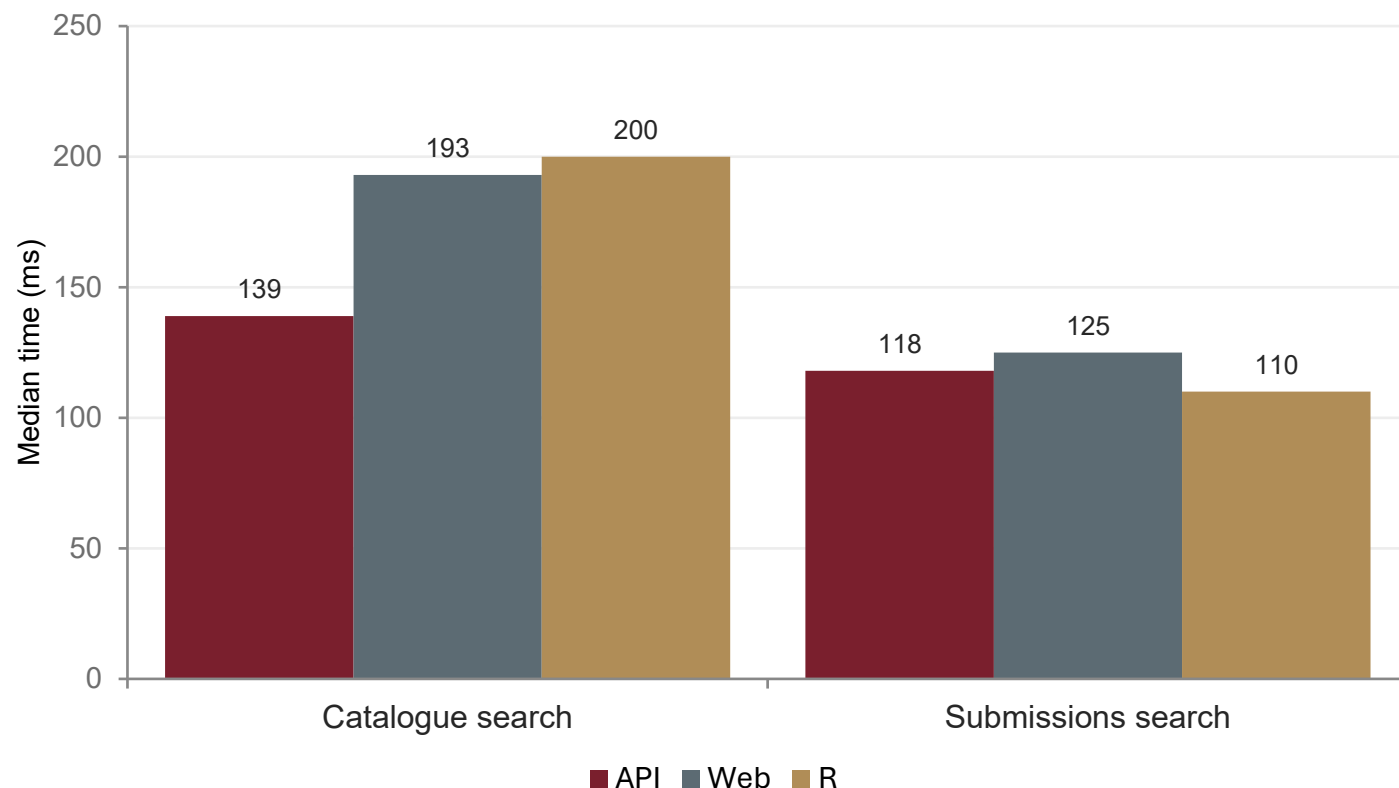
Did we break the website’s answers? No.

Query	Web	API	R
Catalogue: sex = Male	5044	5044	5044
Catalogue: specificepithet = pusillus	126	126	126
Catalogue: Trogon / Male / rufus / Juvenile	2	2	2
Submissions: reference = Zoological Journal of the Linnean Society	1	1	1
Submissions: lastname = Bitton	2	2	2

11 queries checked in the thesis - including thousands of rows - **0 disagreements** between web, API, and R.

Supports the core claim: the API and R client reuse existing web behaviour, rather than producing a separate result set. This is not exhaustive verification of every filter combination.

Response times: indicative, not a benchmark



- Medians ~110–200 ms across both queries and all three interfaces
- /ping baseline ~156 ms - network round-trip is a big slice of the total
- Three runs per condition, one client machine - suitable for interactive scripts, not a formal benchmark

Limitations and future work

LIMITATIONS

- Open /api/v1, with no keys or rate limits, matching public web
- No pagination; large downloads can be slow
- Parity checks are not exhaustive; no external user study
- Package on GitHub, not yet on CRAN

FUTURE WORK

- Optional API keys / rate limiting
- CI-integrated regression parity set
- OpenAPI specification + CRAN submission path
- Controlled benchmarks; pagination and /api/v2

Contributions

Versioned REST API

/api/v1 mirroring Nature's Palette search and download behaviour

naturepaletteR

An R package enabling scripted workflows for spectral data research

Evaluation

Automated tests plus documented parity and performance evaluation

A browser-only repository → reproducible, programmatic access (the website left intact).

Thank you

Questions welcome

Production API

<http://134.87.9.138/api/v1>

R package

github.com/tpturjo/naturepaletteR

Supervisors

Dr. Adrian Fiech · Dr. Pierre-Paul Bitton